

HEFEIFEI JIANG

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RIVERSIDE, CA

EDUCATION

University of California, Riverside (UCR)

M.S. | *Robotics*

Sep 2025 - Present

Riverside, CA, USA

Shaanxi University of Science and Technology (SUST)

B.Eng. (First-class degree) | *Computer Science and Technology*

Sep 2020 - Jun 2024

Xi'an, China

Joint-Degree Undergraduate Program with Ulster University, UK

RESEARCH INTEREST

My research centers on **human-robot interaction**: how robots and people can work together effectively in shared physical spaces. I am interested in robots that involve humans in the loop, whether by taking guidance and corrections from people, assisting and following them, or adapting to their individual preferences, and in what makes such interaction natural, safe, and low-effort for the human. I approach these questions through **robot learning** and **manipulation**, using learning-based methods to enable robots to act with and around people. My goal is to build interactive robots that people find intuitive to guide, trust, and collaborate with.

PUBLICATIONS & PREPRINTS

* Equal contribution.

[6] Mingxuan Yan, Litian Gong, Dihong Huang, Zhixu Li, Tianyu Zhang, Wenqian Zhang, **Hefeifei Jiang**, Peihao Li, Jian Zhang, Mengfei Zhao, Yikai Tang, Hai Zhai, Zehao Wang, Ruijian Liang, Yanjia Huang, Lin Shao, Changliu Liu, Kaiyu Hang, Zhiwen Fan, Masayoshi Tomizuka, Jianfei Yang, Jiachen Li, “What Matters in Quality-Aware VLA Pretraining: A Benchmark and Empirical Study”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. **Under review**.

[5] Shiting Gong, Jianpeng Yao, Litian Gong, **Hefeifei Jiang**, Huaide Jiang, Jinfeng Wang, Zheyu Lin, Mingyu Ding, Jiachen Li, “Closed-Loop Customizable Human Following via Distilling Monotonic Constrained Policies”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. **Under review**.

[4] Zehao Wang*, **Hefeifei Jiang***, Siyan Li, Guoyuan Wu, Jiachen Li, “CoopLight: Cooperative Vision-Language Agents for Long-Tail Traffic Control”, *International Conference on Learning Representations (ICLR)*, 2027. **Under review**.

[3] Zhefei Gong*, **Hefeifei Jiang***, Zehao Wang, Jiachen Li, “HiRe-MoMa: Hierarchical Residual Policy Learning for Real-World Mobile Manipulation with Sparse Human Guidance”, *Conference on Robot Learning (CoRL)*, 2026.

[2] Y. Qi, M. Qiu, **H. Jiang**, F. Wang, “Extracting Fingerprint Features Using Autoencoder Networks for Gender Classification”, *Applied Sciences*, 12(19): 10152, 2022.

[1] Y. Qi, **H. Jiang**, S. Li, J. Cao, “ConvLSTM Coupled Economics Indicators Quantitative Trading Decision Model”, *Symmetry*, 14(9): 1896, 2022.

PATENTS

[2] Y. Qi, **H. Jiang**, F. Wang, J. Yang, “A Method and Apparatus for Matching a Three-dimensional Model of the Sacrum Based on Deep Learning”, Chinese patent.

[1] Y. Qi, **H. Jiang**, X. Jin, J. Ma, M. Zhao, Y. Liu, K. Qian, Y. Wang, Z. Wang, H. Cui, “AIOT Digital Twin Intelligent Substation Cloud-based Analysis Platform”, Chinese patent.

RESEARCH EXPERIENCE

Trustworthy Autonomous Systems Lab (TASL)

Advisor: Prof. Jiachen Li, UC Riverside

2025 - Present

Riverside, CA, USA

- **HiRe-MoMa: Real-World Fine-Tuning of Mobile Manipulation with Sparse Human Guidance (Co-first author)**
 - ▶ Pre-trained visuomotor policies often fail on real mobile manipulators, and fine-tuning them on hardware is costly because base and arm errors compound.
 - ▶ Proposed a hierarchical residual RL framework that learns separate base and arm corrections on top of a frozen base policy, guided by sparse human interventions through expert-guided value alignment.

- ▶ On a TIAGo++, raised placement success from 50% to 95% with about 30 min of online interaction and <20% human intervention.
- **CoopLight: Cooperative Vision-Language Agents for Long-Tail Traffic Control (Co-first author)**
 - ▶ Traffic signals and connected vehicles are usually controlled separately from low-dimensional states, which breaks down in long-tail events such as incidents, jaywalking, and emergency vehicles.
 - ▶ Proposed a cooperative vision-language framework in which signal and vehicle agents answer agent-specific VQA over multi-view images and exchange semantic reports for joint decisions.
 - ▶ In closed-loop SUMO and Blender simulation of two real intersections, cut average waiting time by 26% overall and 56% for connected vehicles.
- **Closed-Loop Customizable Human Following via Distilling Monotonic Constrained Policies (Co-author)**
 - ▶ Standard RL human-following policies cannot change the preferred following distance without retraining.
 - ▶ Deployed the distilled policy on a ROSMASTER X3 with LiDAR-based pedestrian tracking, ran real-world experiments in an open area, a corridor, and a cafe, and recorded the demonstration videos.
 - ▶ Showed sim-to-real transfer with test-time distance adjustment and no policy fine-tuning.
- **AXIS-Bench: Quality-Aware VLA Pretraining from Suboptimal Human Demonstrations (Co-author)**
 - ▶ Human demonstrations for VLA pretraining vary in quality, with hesitations and detours.
 - ▶ Ran the real-world evaluation on a Franka FR3 with a Robotiq gripper and ZED cameras, testing fine-tuned $\pi_{0.5}$ policies with and without simulation pretraining across ten target tasks.
 - ▶ Showed that simulation pretraining raises real-world success from 9.6% to 31.7%, improving nine of ten tasks.

Quantum Neural Networks for Efficient Multimodal Representation Learning

Jun 2024 - Dec 2024

Remote Research Intern, Advisor: Prof. Juan Zhuang

Remote

- Replaced the CLIP image encoder with a Quantum Neural Network, reaching competitive accuracy on benchmark datasets with a substantially reduced parameter budget.

Summer Workshop at National University of Singapore

Jun 2024 - Jul 2024

Team Leader, Advisor: Prof. Tan Wee Kek

Singapore

- Led a multinational team building a greenhouse monitoring system with Raspberry Pi, ESP32 and multi-modal sensors, with a web app for real-time monitoring.
- Applied YOLOv8 for plant counting and a ResNet classifier for pest and disease recognition on live video streams.

Digital Twin Intelligent Grid Physical Fusion Platform

Apr 2021

Robot Embedded Development

China

- Built a tracked substation inspection robot with SLAM mapping, obstacle detection and a YOLOv3 recognition module; deployed in multiple substations as a provincial-level innovation project.

SKILLS

Programming & ML: Python, C/C++, PyTorch, Bash, Git, LaTeX

Robotics: ROS, TIAGo++, Franka, Diffusion Policy, ACT

Simulation & Benchmarks: IsaacLab, DROID, LIBERO, SUMO, Webots

Methods: Reinforcement Learning, Imitation Learning, Residual Policy Learning, Vision-Language Models

HONORS & AWARDS

International

- **Meritorious Award**, Mathematical Contest in Modeling (MCM) Feb 2022
- **Honorable Mention**, ICPC Asia-East Continent Final Apr 2021
- **Gold Medal**, International Festival of Science and Technology, Tunisia Aug 2020

Domestic

- **"Zhi Cheng Zhi Bo" All-Round Excellence Scholarship (Top 10)**, University-level Scholarship Jun 2024
- **Zhou Lian Scholarship (Top 10, #1 by points)**, University-level Scholarship Apr 2024
- **5th Place**, 4th "Keda" Cup Five-a-Side Football Tournament Jun 2024
- **3rd Prize**, National College Optoelectronic Design Competition Aug 2022
- **3rd Prize**, National College Computer Design Competition Aug 2022
- **Ranked 215 / 627**, WeChat Big Data Challenge May 2022
- **Excellence Award**, China AI+ Innovation and Entrepreneurship Competition Dec 2021
- **Provincial-Level Approval**, China College Student Innovation and Entrepreneurship Project Nov 2021
- **1st Prize**, 23rd National Robot Championship & 12th International Humanoid Robot Olympics Oct 2021
- **3rd Prize**, National College Computer Design Competition Aug 2021